1. Difference Between Live Connection and Extract Connection in Tableau

Live Connection: A live connection directly connects Tableau to the original data source. When you interact with a visualization, Tableau queries the data source to retrieve the latest available data.

Extract Connection: An extract connection creates a copy of the data and stores it as a Tableau extract (.hyper file). Tableau uses this stored data to create visualizations, which can improve performance.

Basis	Live Connection	Extract Connection
Data storage	Data remains in the original source.	Data is copied and stored as an extract.
Data updates	Queries the source for the latest available data.	Data updates when the extract is refreshed.
Performance	Depends on the source and network speed.	Often faster because data is stored locally or in Tableau's environment.
Internet/network	Usually requires access to the data source.	Can work without access to the original source after the extract is created, if the extract is accessible.
Best suited for	Data that needs to remain up to date.	Faster analysis and working with a stored snapshot of data.

Example:

- Live: Tableau connects directly to a company's database to retrieve current sales data.
- Extract: Tableau stores a copy of the sales data and uses it for analysis.

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2. Difference Between Data Blending and Joining in Tableau

Data Joining: Joining combines data from two or more tables based on a common field. The tables are combined at the row level using join types such as Inner Join, Left Join, Right Join, and Full Outer Join.

Data Blending: Data blending combines data from different data sources in a worksheet. One data source acts as the primary source, while another acts as the secondary source. Tableau links the data using common fields and aggregates the secondary data as needed.

Basis	Data Joining	Data Blending
Meaning	Combines tables into a single logical table.	Combines data from different sources in a visualization.
Data sources	Commonly used with tables from the same data source.	Designed to combine data from different data sources.

Basis	Data Joining	Data Blending
How data is combined	Uses join conditions and matching fields.	Uses relationships between common fields to combine aggregated results.
Level of combination	Row level.	Aggregated level.
Join types	Supports Inner, Left, Right, and Full Outer Joins.	Does not use traditional join types.
Data handling	Can produce duplicate rows if the join keys are not unique.	Aggregation and linking behavior can help avoid some row-level duplication issues.
Example	Joining Customer and Orders tables using Customer ID.	Comparing sales data from Excel with target data from another data source.

Example:

- **Joining:** Combining Customer and Orders tables using Customer ID to create a combined table.
- **Blending:** Comparing actual sales from an Excel file with sales targets from another data source in the same Tableau visualization.